信号与系统

School of Electronic
and Information Engineering
Zhou Hang

2026.3

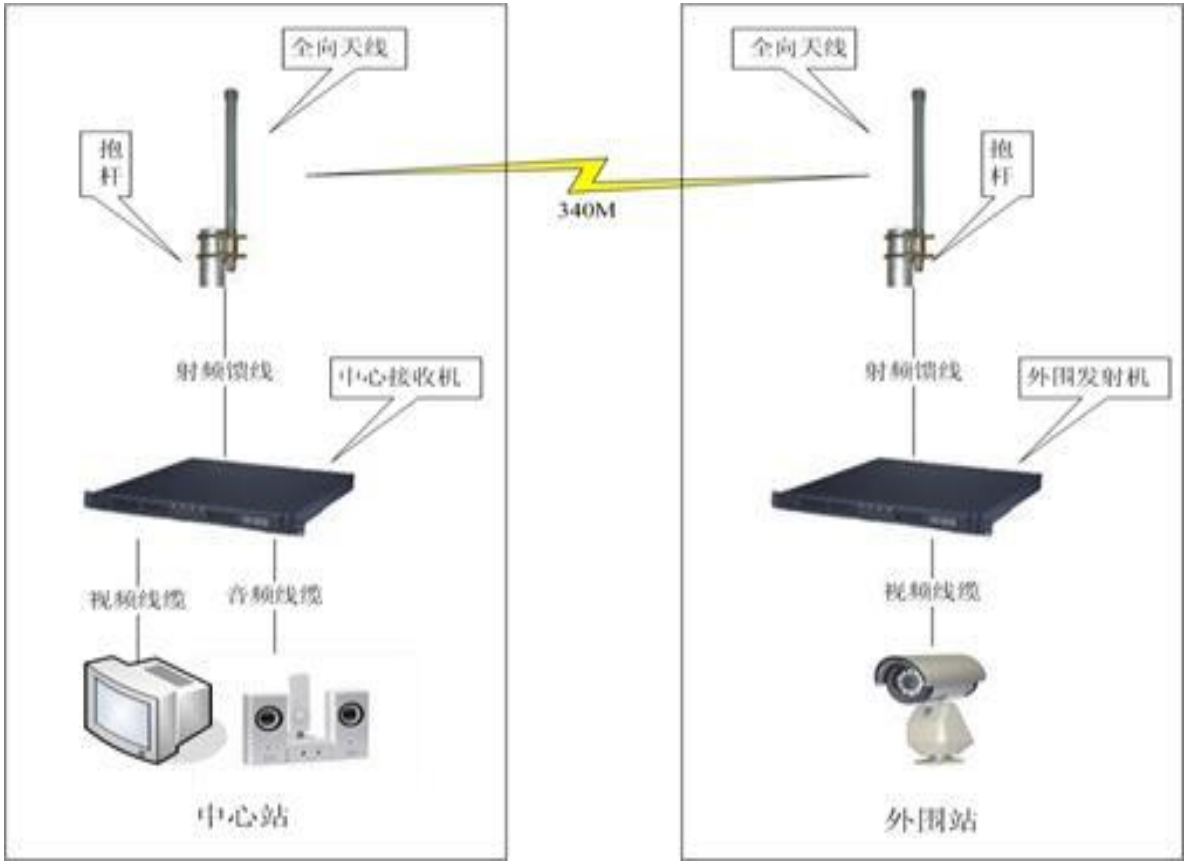
1. Introduction to Signal and System Analysis

- Introduction
- Signal description and classification
- System description and classification
- Applications of signals and systems

Applications of signals and systems

- communication field
- control area
- signal processing
- biomedical science

1.Applications of signals and systems in the field of communication



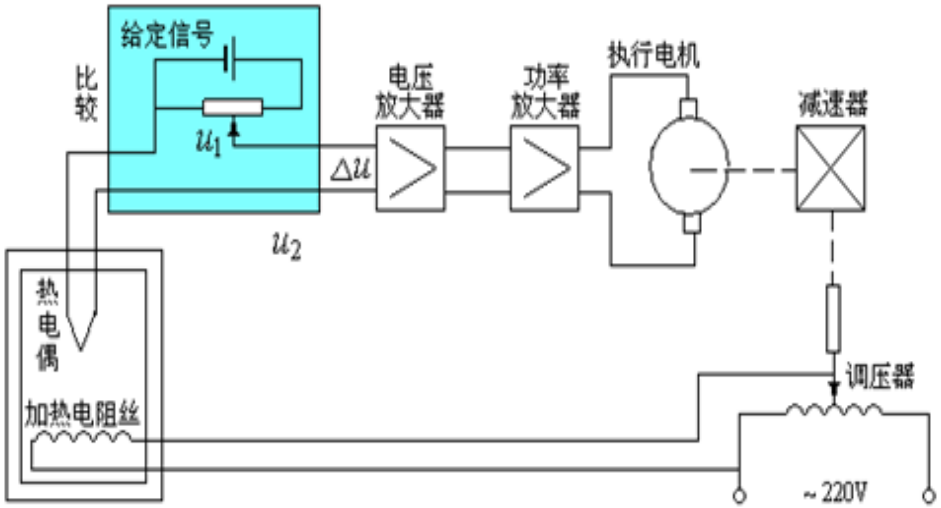
※How is the signal transmitted?

※How to improve transmission quality, transmission efficiency and security.

broadcast image communication system

2.Applications of signals and systems in the field of control

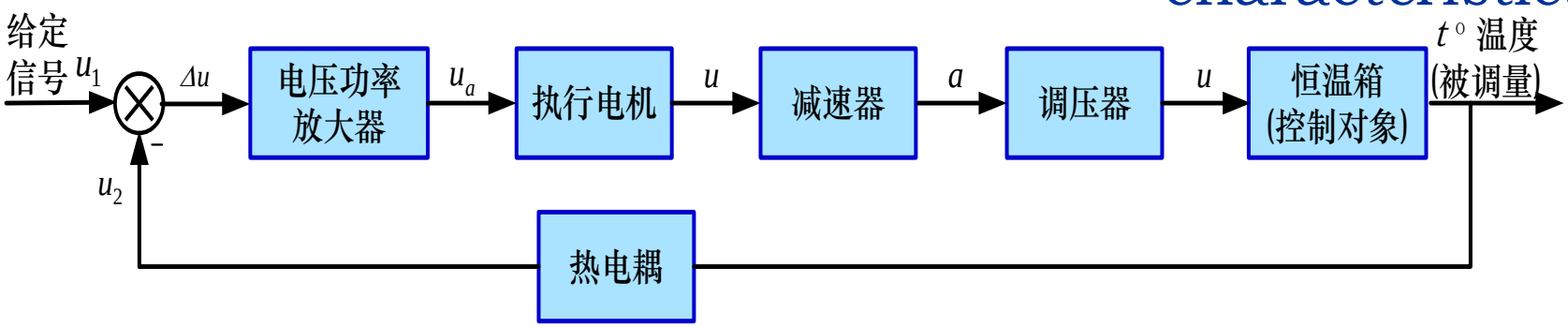
恒温箱



恒温箱自动控制系统

※How to analyze system transmission characteristics and stability?

※How to adjust the desired system characteristics



3.Applications of Signals and Systems in Signal Processing

➢ Signal characteristic analysis

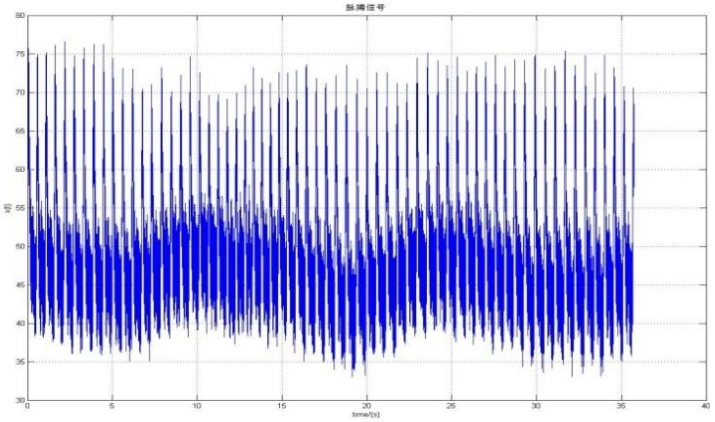
Pulse sensor

Signal processing circuit

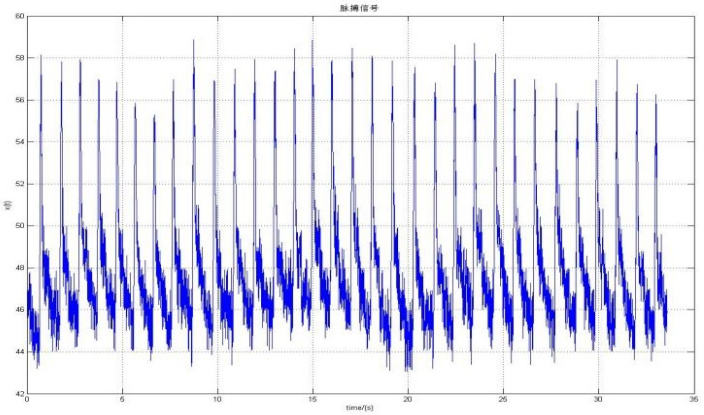
BJTU-DSP5502

Computer

Pulse signal acquisition system



waking state pulse signal



Pulse signal in fatigue state

3.Applications of Signals and Systems in Signal Processing

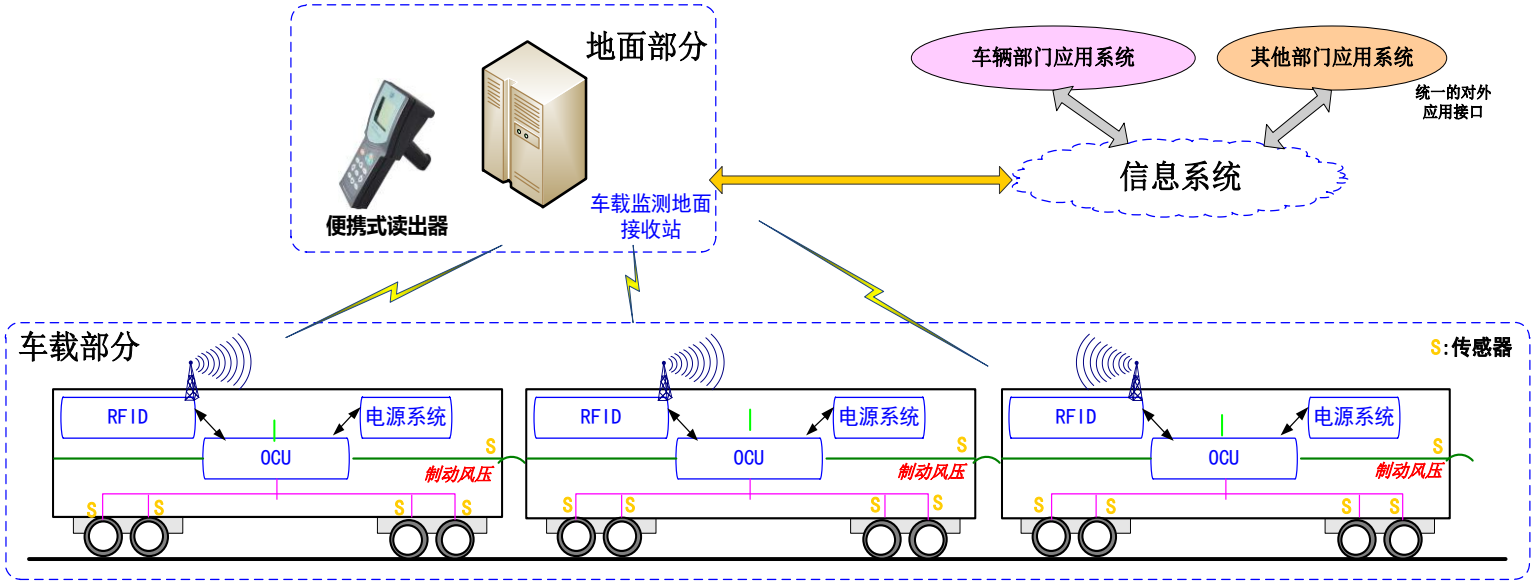
➢ Signal characteristic analysis

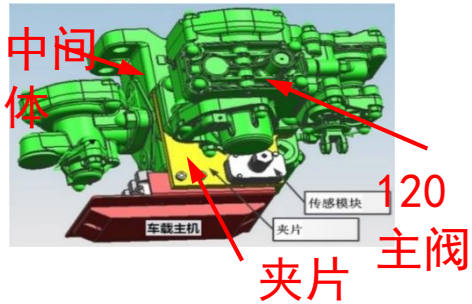
On-board safety monitoring of trucks

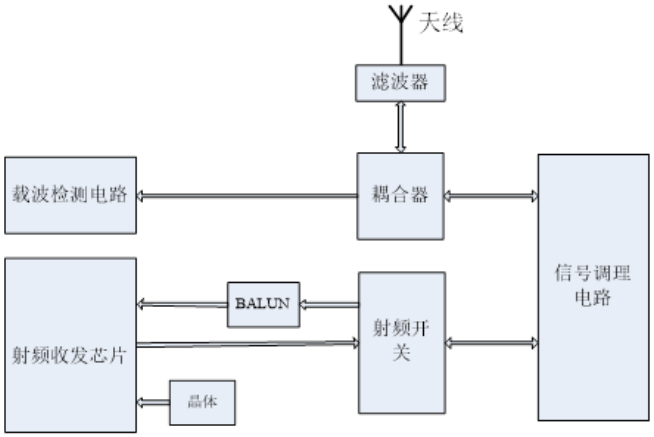
3.Applications of Signals and Systems in Signal Processing

➢ Signal characteristic analysis

On-board safety monitoring of trucks



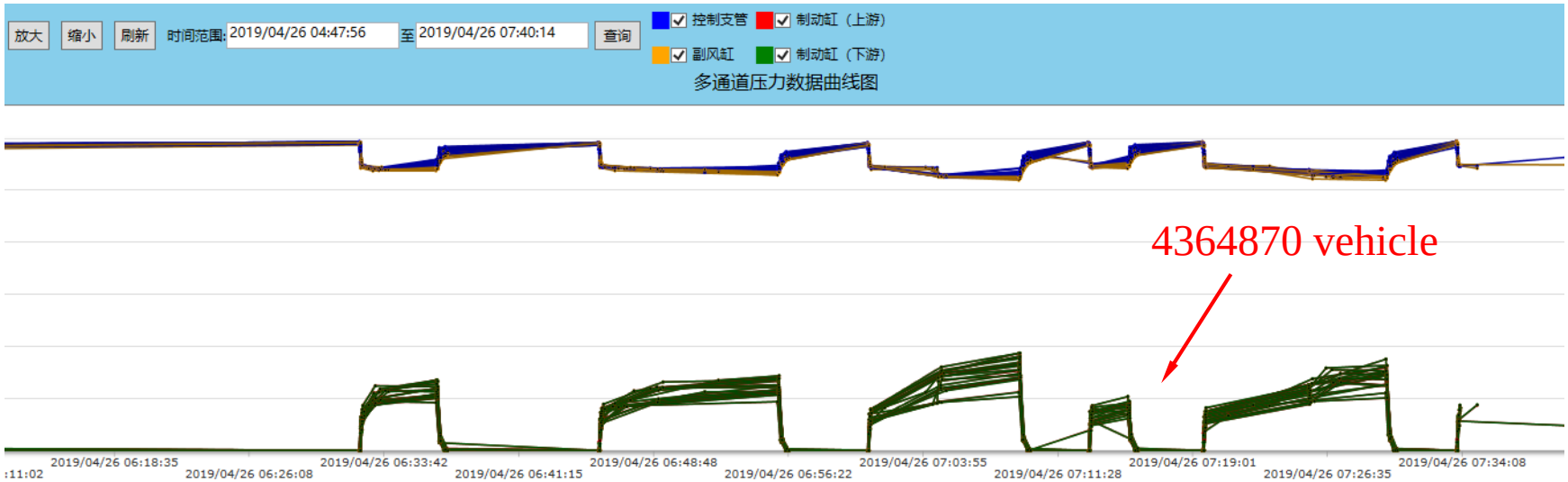




3.Applications of Signals and Systems in Signal Processing

During the line operation test of the on-board safety monitoring system for trucks on the Daqin Railway, the project team analyzed the test data and found that some vehicles had abnormal braking data. Relevant maintenance has been carried out, which has reduced related risks to a certain extent and increased the safety of vehicle operation.

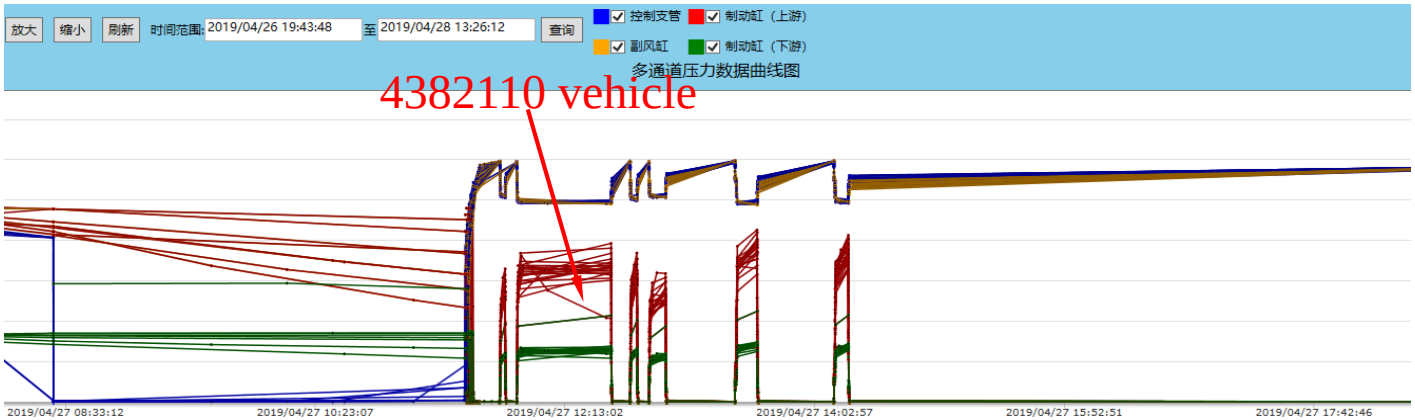
(1) It was found that 4364870 vehicle had natural mitigation failures many times



On April 26, 4364870 vehicle Cha Wu abnormal data

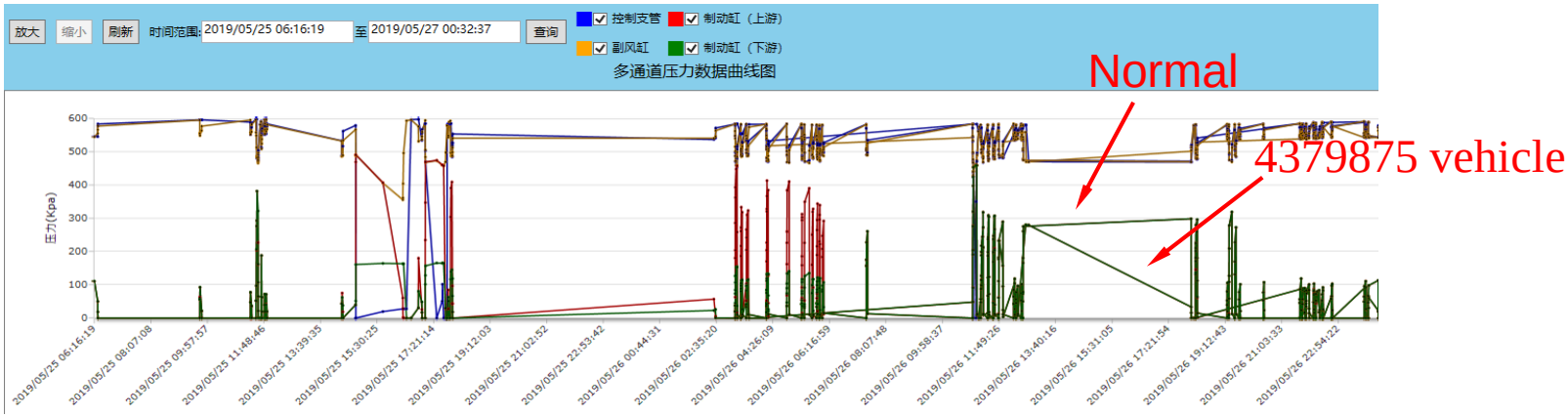
3.Applications of Signals and Systems in Signal Processing

(2) It was found that the 4382110 vehicle had multiple leakage failures



On April 26, 4382110 vehicle Zhu Lu abnormal data

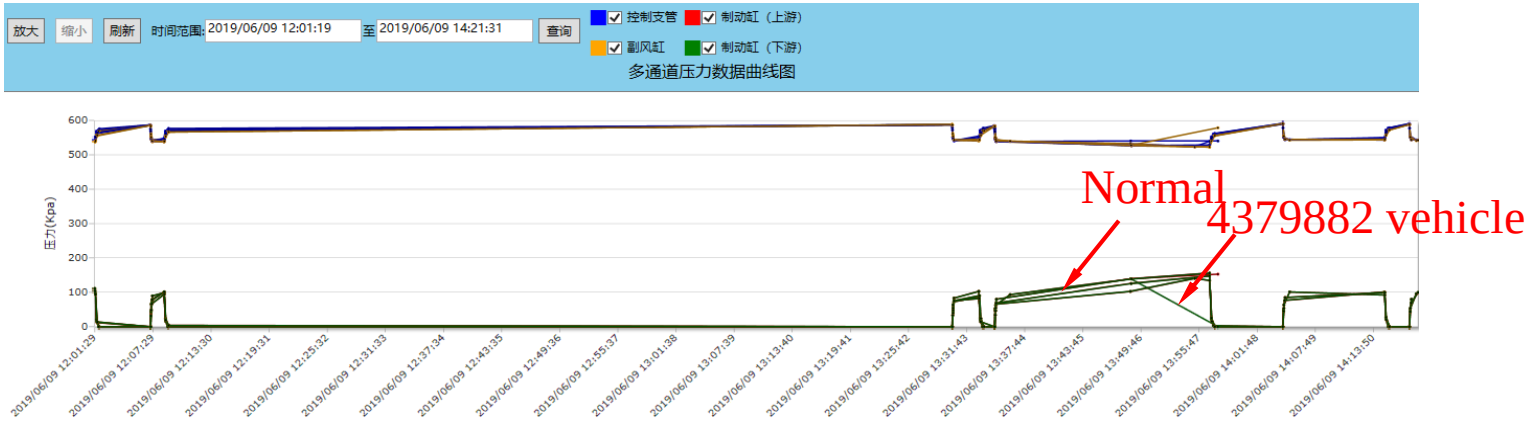
(3) 4379875 vehicles found natural mitigation and leak failures



4379875 vehicle Zhu Lu abnormal data

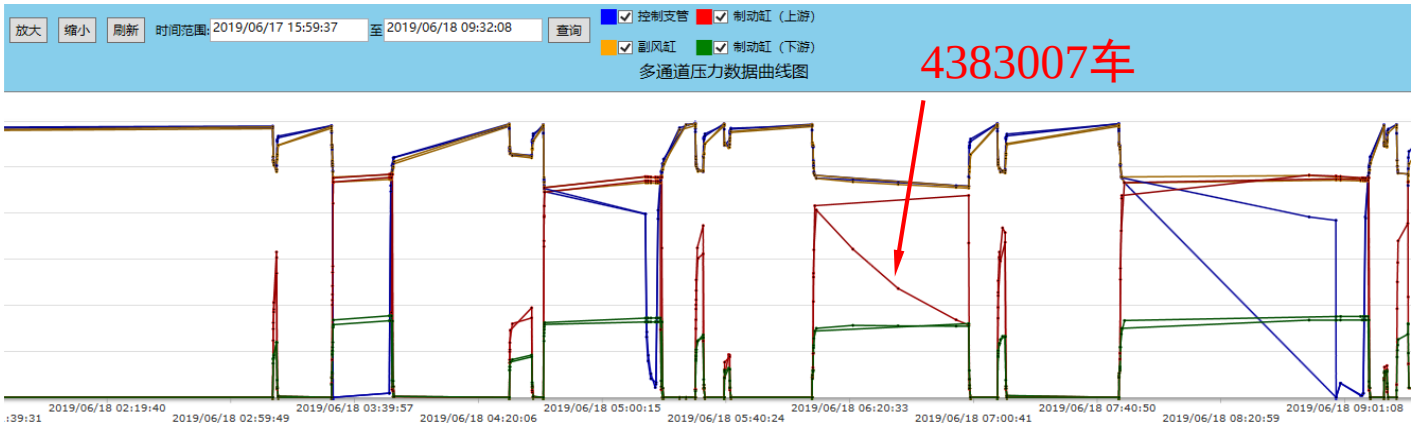
3.Applications of Signals and Systems in Signal Processing

(4) 4379882 vehicles were found to have natural mitigation failures



4379882 vehicle Zhu Lu abnormal data

(5) It was found that the 4383007 car had leaked and failed many times

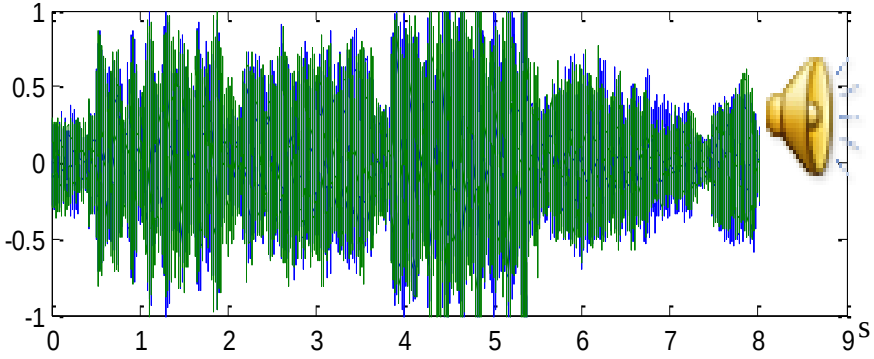


On June 18, 4383007 vehicle Zhu Lu abnormal data

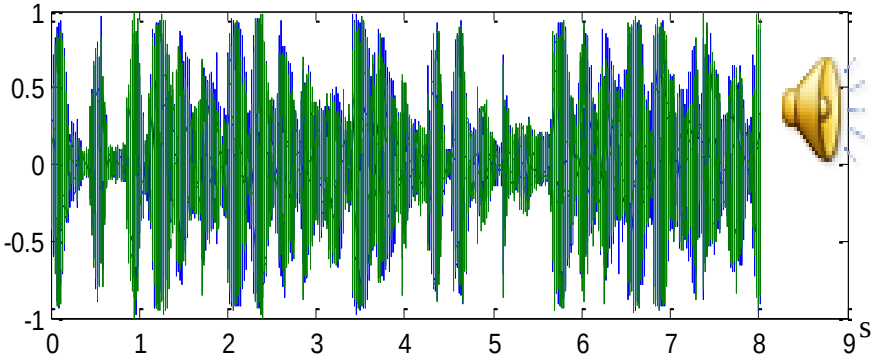
3.Applications of Signals and Systems in Signal Processing

➢ Signal characteristic analysis

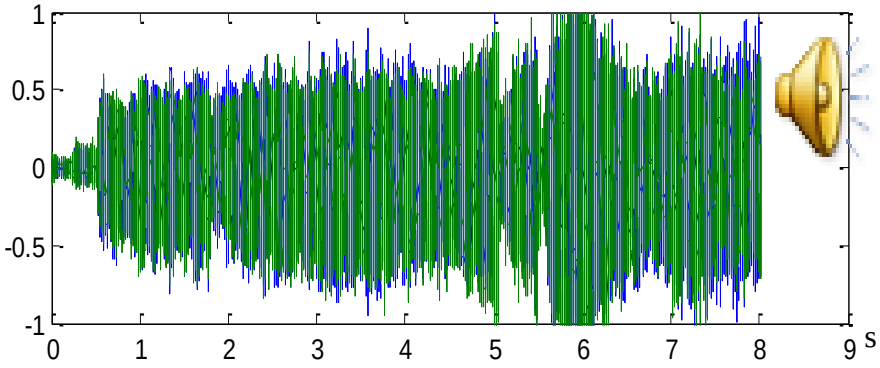
我和你



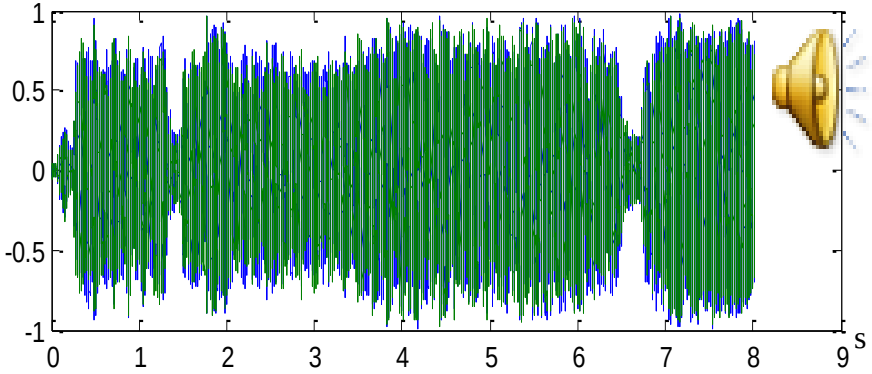
走天涯



我的太阳

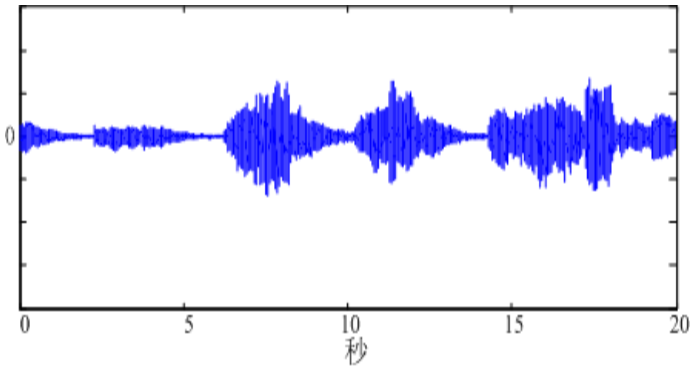


我的太阳

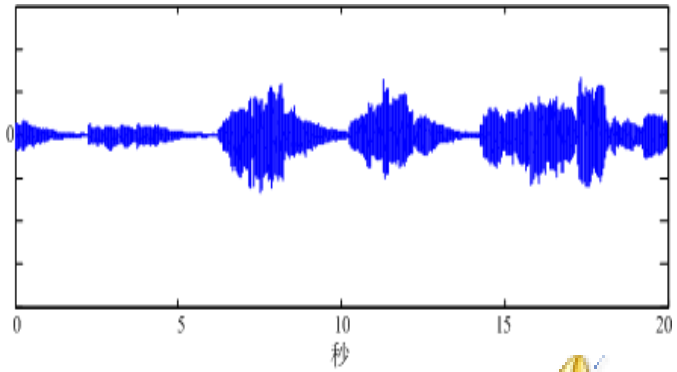


3.Applications of Signals and Systems in Signal Processing

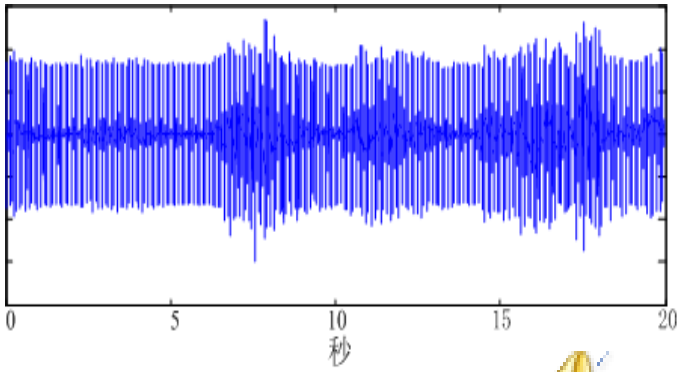
➢ Noisy signal denoising



Time domain waveform of the signal



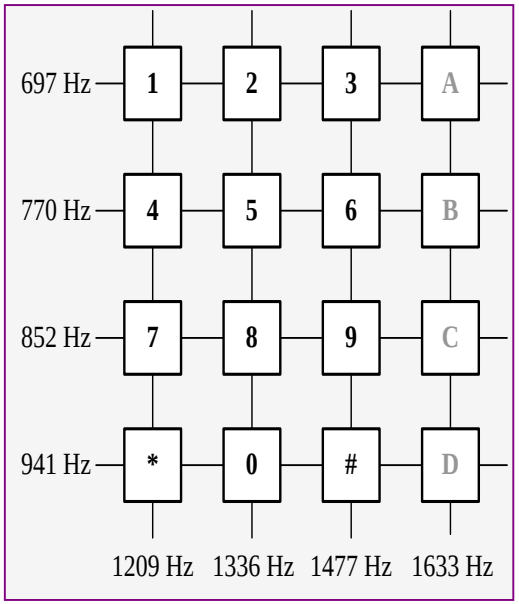
Waveform of filtered signal



Time domain waveform of noise added signal

3.Applications of Signals and Systems in Signal Processing

➢ Signal Recognition - Phone Number Recognition

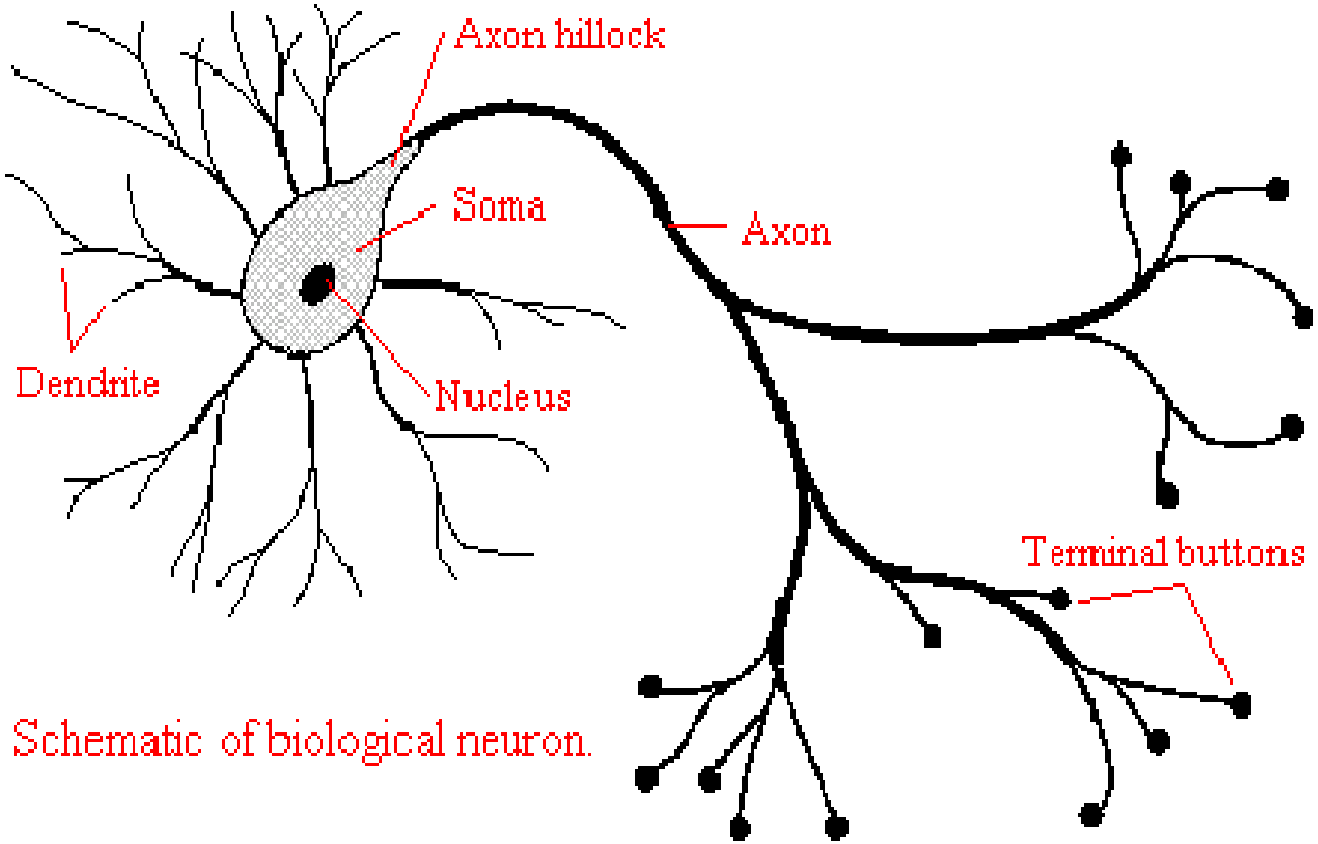


Beijing Jiaotong University
Signal Processing Course
Group telephone dial tone

3.Applications of Signals and Systems in Signal Processing

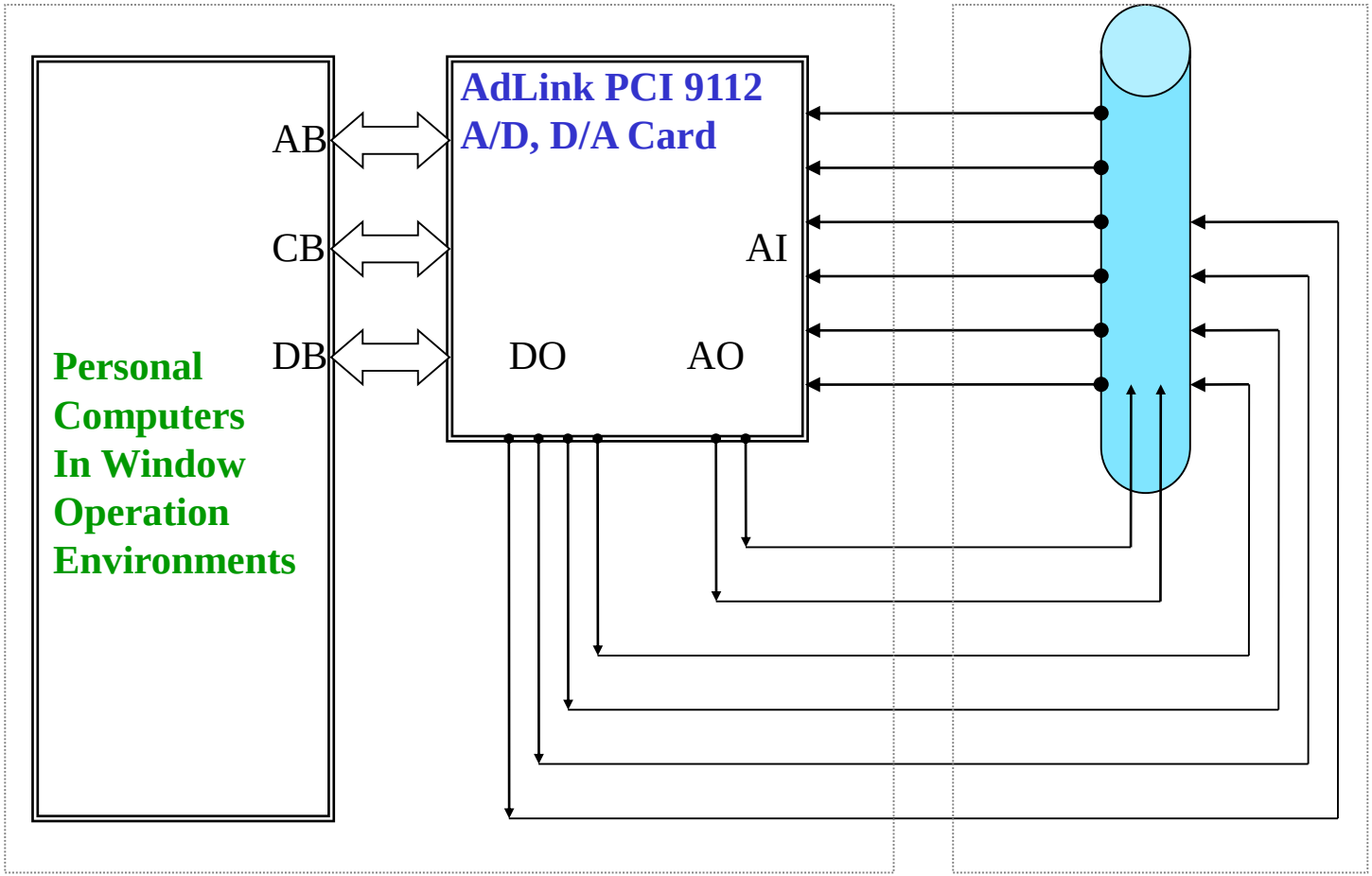
- Signal Recognition - Speaker Recognition

4.Applications of Signals and Systems in Biomedicine



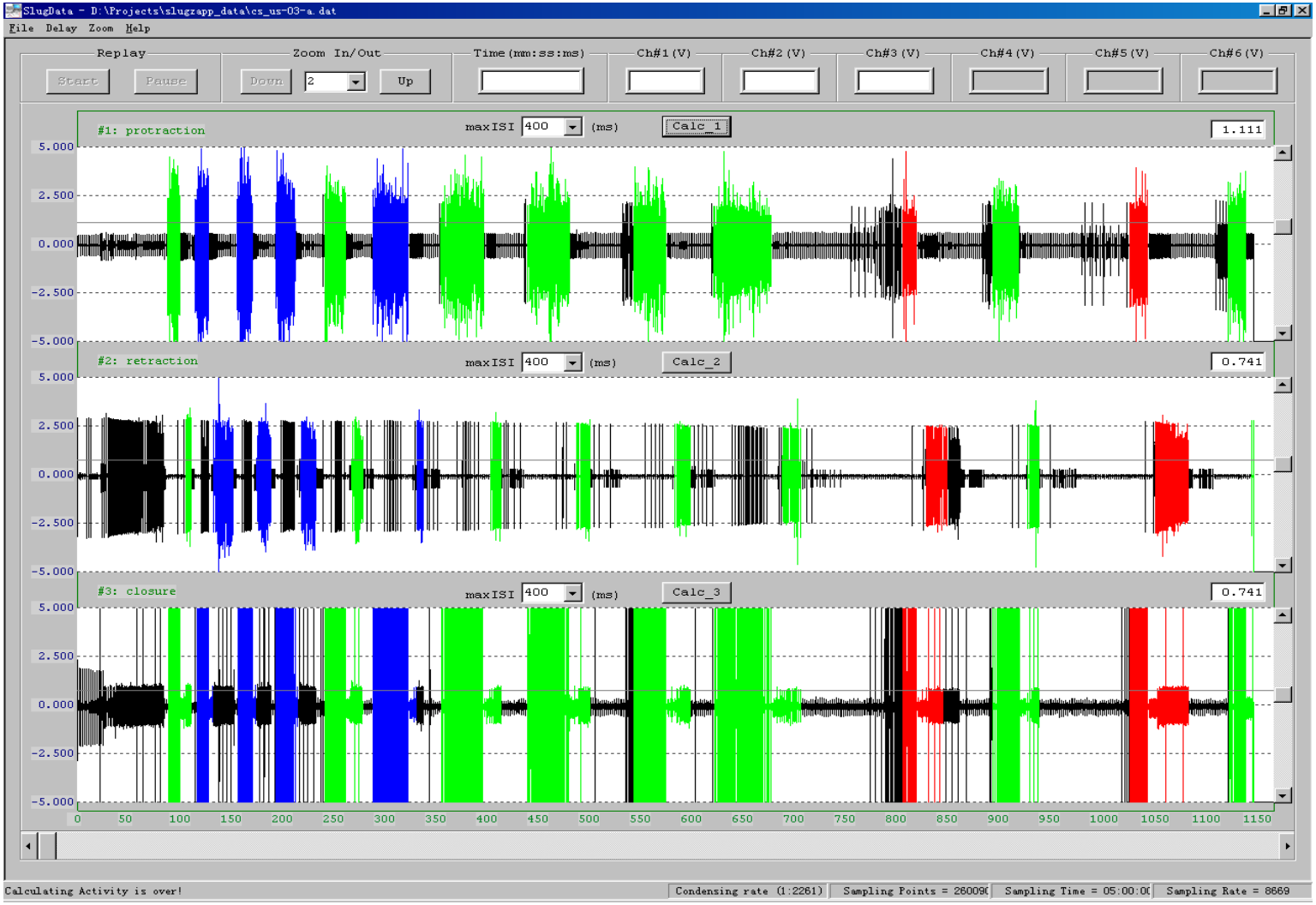
Biological nerve cell (unit) structure diagram

4.Applications of Signals and Systems in Biomedicine



Biosignal Acquisition System

4.Applications of Signals and Systems in Biomedicine



Pattern Recognition of Biosignals

Thanks

Some of the materials quoted in this course are accumulated by the main lecturer over the years, from a variety of media and colleagues, peers and friends of the exchange, it is difficult to attribute them all, so I would like to state and thank you.